

DD_EM-03 - CVM Retention And Recovery

Version: v1.0 **Bundle:** 10_other **Status:** Draft for review **Primary owner:** Customer value management / commercial operations team **Related DDs:** DD_ILM-CFG-01, DD_CUST-INT-01, DD_TCK-01, DD_BIL-04, DD_BIL-01, DD_SUB-LM-01, DD_SUB-WF-UPGRADE-01, DD_SUB-WF-DOWNGRADE-01, DD_SIP-03, DD_SIP-05, DD_DIS-OP-01, DD_NOT-01, DD_REP-01, DD_EM-CFG-03, DD_EM-CFG-04, FOUNDATION_DROOLS

MVP simplification baseline: EM-03 starts as rule-based CVM queues, offers, outcomes, and activity tracking. Do not build an ML platform, campaign automation suite, or separate analytics warehouse for CVM in the first release. Use REP-01/BIL/SUB/TCK inputs and call owning module APIs for actions.

1. Purpose

EM-03 defines customer value management processes for retention, recovery, churn prevention, upsell/cross-sell targeting, win-back, and customer outreach activity management.

Use EM-03 when:

- a customer enters dunning or repeated payment risk;
- a subscription is downgraded, restricted, suspended, cancelled, or at churn risk;
- a customer repeatedly opens support complaints or installation issues;
- commercial teams want retention offers, recovery activities, or win-back campaigns;
- marketing/CVM needs to create targeted customer cohorts and assign outbound activities;
- the system must track offer outcomes and customer interaction history.

EM-03 does not own customer master, billing, subscription state, campaigns, discounts, or notification delivery. It consumes events and calls owning module APIs.

2. Scope And Boundaries

Capability	Owner
CVM segment, customer risk score, activity, offer decision, outcome	EM-03
Customer/account/contact source of truth	ILM-CFG-01
Customer interaction timeline	CUST-INT-01
Ticket/support case history	TCK-01 / ASR flows
Dunning and billing recovery state	BIL-04 / BIL-01
Subscription state and operations	SUB-LM / SUB-WF
Promotions/campaign catalog	SIP-05
Discount assignment and runtime discount application	SIP-03 / DIS-OP-01
Customer messages	NOT-01
Permissions	EM-CFG-03
Approval for exceptional retention offer	EM-CFG-04
Reporting	REP-01
Configurable eligibility, score band, and offer rules	Drools

Boundary rules:

- EM-03 does not directly apply discounts. It calls SIP-03 or campaign/discount APIs.
- EM-03 does not directly resume, restrict, or cancel subscriptions. It calls SUB/BIL workflows.
- EM-03 does not send SMS/email directly. It requests NOT-01 notification.
- EM-03 writes interaction/activity events to CUST-INT-01.
- EM-CFG-04 is used for non-standard retention offers, manual overrides, or high-value discounts.

- Camunda is involved only when the approval policy mode in EM-CFG-04 is BPMN.

3. Process Overview

```
flowchart LR
    A["BIL/SUB/TCK/customer events"] --> B["EM-03 updates signal profile"]
    B --> C["Drools evaluates segment and risk"]
    C --> D["Create CVM activity or offer"]
    D --> E["Assign user/team or trigger notification"]
    E --> F["Customer accepts, rejects, or no response"]
    F --> G["Call owning module API"]
    G --> H["Record outcome and interaction"]
```

4. Data Model

4.1 cvm_customer_signal_profile

What it is for: stores aggregated customer risk and opportunity signals used by CVM.

When to use it: update this profile from billing, subscription, ticket, campaign, payment, and interaction events.

Sample meaning: csp-001 says customer CUS-100045 has medium churn risk due to repeated dunning and complaints.

Column	Type	Purpose	Sample
signal_profile_id	TEXT PK	Profile id.	csp-001
operator_code	TEXT	Tenant/operator.	WIK
customer_id	TEXT	Customer id.	CUS-100045
account_id	TEXT NULL	Account id.	ACC-900045
active_subscription_count	INT	Active services.	1
open_ticket_count	INT	Open tickets.	2
dunning_level	INT NULL	Current dunning level.	2
days_since_last_payment	INT NULL	Payment recency.	18
complaint_count_90d	INT	Recent complaint count.	3
churn_risk_score	DECIMAL	0-100 risk score.	72.5
upsell_score	DECIMAL	0-100 upsell score.	38.0
last_evaluated_at	TIMESTAMP	Last rule/scoring time.	2026-08-25T08:00:00Z

Recommended indexes:

- unique (operator_code, customer_id).
- (operator_code, churn_risk_score).
- (operator_code, dunning_level).

4.2 cvm_segment_membership

What it is for: stores current CVM segment membership for targeting and reporting.

When to use it: create/update rows when Drools places a customer into retention, recovery, win-back, upsell, or VIP-care segments.

Sample meaning: csm-001 places customer CUS-100045 into RETENTION_HIGH_RISK.

Column	Type	Purpose	Sample
membership_id	TEXT PK	Membership id.	csm-001
operator_code	TEXT	Tenant/operator.	WIK
segment_code	TEXT	Segment code.	RETENTION_HIGH_RISK
customer_id	TEXT	Customer id.	CUS-100045
status	TEXT	ACTIVE, EXPIRED, SUPPRESSED.	ACTIVE
reason_code	TEXT	Why customer is in segment.	DUNNING_AND_COMPLAINTS

Column	Type	Purpose	Sample
entered_at	TIMESTAMP	Entry time.	2026-08-25T08:00:00Z
expires_at	TIMESTAMP NULL	Expiry time.	2026-09-25T08:00:00Z

Recommended indexes:

- (operator_code, segment_code, status).
- (customer_id, status).

4.3 cvm_activity

What it is for: stores a customer outreach, retention, recovery, upsell, or win-back activity.

When to use it: create one row when a user/team or automated process must contact a customer or propose an offer.

Sample meaning: cva-001 is a recovery call assigned to the retention team for customer CUS-100045.

Column	Type	Purpose	Sample
activity_id	TEXT PK	Activity id.	cva-001
operator_code	TEXT	Tenant/operator.	WIK
activity_type	TEXT	RETENTION_CALL, PAYMENT_RECOVERY, UPSELL_OFFER, WINBACK, SERVICE_RECOVERY.	PAYMENT_RECOVERY
customer_id	TEXT	Customer id.	CUS-100045
account_id	TEXT NULL	Account id.	ACC-900045
subscription_id	TEXT NULL	Subscription context.	SUB-700045
source_event_ref	TEXT NULL	Trigger event/case ref.	dunning-evt-001
assigned_to_user_id	TEXT NULL	Owner user.	usr-retention-01
assigned_team_id	TEXT NULL	Owner team.	team-retention
status	TEXT	OPEN, IN_PROGRESS, WAITING_CUSTOMER, COMPLETED, CANCELLED, EXPIRED.	OPEN
priority	TEXT	LOW, MEDIUM, HIGH, CRITICAL.	HIGH
due_at	TIMESTAMP NULL	Due time.	2026-08-26T17:00:00Z
created_at	TIMESTAMP	Creation timestamp.	2026-08-25T08:05:00Z
closed_at	TIMESTAMP NULL	Close timestamp.	null

Recommended indexes:

- (operator_code, status, priority).
- (assigned_to_user_id, status).
- (customer_id, status).

4.4 cvm_offer_instance

What it is for: stores one offer proposed to a customer through CVM.

When to use it: create one row before presenting a discount, campaign, payment plan referral, upgrade, recovery credit, or service-goodwill offer.

Sample meaning: cvo-001 proposes a 10% three-month retention discount to customer CUS-100045.

Column	Type	Purpose	Sample
offer_instance_id	TEXT PK	Offer instance id.	cvo-001
operator_code	TEXT	Tenant/operator.	WIK
activity_id	TEXT NULL	Parent activity.	cva-001
customer_id	TEXT	Customer id.	CUS-100045
subscription_id	TEXT NULL	Subscription context.	SUB-700045

Column	Type	Purpose	Sample
offer_type	TEXT	RETENTION_DISCOUNT, UPGRADE_OFFER, WINBACK_PACKAGE, GOODWILL_CREDIT, PAYMENT_REMINDER.	RETENTION_DISCOUNT
campaign_code	TEXT NULL	SIP-05 campaign.	RET-10PCT-3M
discount_ref	TEXT NULL	Discount/catalog ref.	DISC-RET-10
status	TEXT	DRAFT, PROPOSED, ACCEPTED, REJECTED, EXPIRED, APPLIED, FAILED.	PROPOSED
approval_request_id	TEXT NULL	Approval ref.	apr-cvm-001
expires_at	TIMESTAMP NULL	Offer expiry.	2026-08-31T23:59:59Z
created_at	TIMESTAMP	Creation timestamp.	2026-08-25T08:10:00Z

Recommended indexes:

- (operator_code, status).
- (customer_id, status).
- (approval_request_id).

4.5 cvm_outcome

What it is for: stores final outcome of an activity or offer.

When to use it: create/update when customer accepted, rejected, ignored, paid, upgraded, churned, recovered, or when an activity was closed.

Sample meaning: cv00-001 records that customer accepted and discount assignment was created.

Column	Type	Purpose	Sample
outcome_id	TEXT PK	Outcome id.	cv00-001
operator_code	TEXT	Tenant/operator.	WIK
activity_id	TEXT NULL	Activity id.	cva-001
offer_instance_id	TEXT NULL	Offer id.	cv0-001
customer_id	TEXT	Customer id.	CUS-100045
outcome_code	TEXT	ACCEPTED, REJECTED, NO_RESPONSE, PAID, UPGRADED, RECOVERED, CHURNED.	ACCEPTED
owning_module_ref_type	TEXT NULL	Applied module.	DISCOUNT_ASSIGNMENT
owning_module_ref_id	TEXT NULL	Result id.	da-001
notes	TEXT NULL	Closure note.	Customer accepted retention discount.
created_at	TIMESTAMP	Outcome time.	2026-08-25T09:00:00Z

Recommended indexes:

- (customer_id, created_at).
- (outcome_code, created_at).

5. APIs

5.1 Evaluate Customer

POST /api/cvm/customers/CUS-100045/evaluate
Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "reason": "DUNNING_EVENT",
  "sourceEventRef": "dunning-evt-001"
}
```

5.2 Create Activity

```
POST /api/cvm/activities
Idempotency-Key: cvm-dunning-evt-001
Content-Type: application/json

{
  "operatorCode": "WIK",
  "activityType": "PAYMENT_RECOVERY",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "sourceEventRef": "dunning-evt-001",
  "assignedTeamId": "team-retention",
  "priority": "HIGH"
}
```

5.3 Propose Offer

```
POST /api/cvm/offers
Content-Type: application/json

{
  "operatorCode": "WIK",
  "activityId": "cva-001",
  "customerId": "CUS-100045",
  "subscriptionId": "SUB-700045",
  "offerType": "RETENTION_DISCOUNT",
  "campaignCode": "RET-10PCT-3M",
  "discountRef": "DISC-RET-10"
}
```

5.4 Accept Offer

```
POST /api/cvm/offers/cvo-001/accept
Content-Type: application/json

{
  "acceptedByUserId": "usr-retention-01",
  "customerConsentRef": "consent-001"
}
```

EM-03 then calls the owning module, such as SIP-03 discount assignment or SUB-WF upgrade request.

6. Rules And Drools

Recommended packages:

- rules.cvm.segmentation
- rules.cvm.activity
- rules.cvm.offer
- rules.cvm.approval

```
package rules.cvm.segmentation

rule "High risk dunning customer"
when
  $f : CvmCustomerFact(dunningLevel >= 2,
                      complaintCount90d >= 2)
  $d : CvmDecision()
then
  $d.addSegment("RETENTION_HIGH_RISK");
  $d.createActivity("PAYMENT_RECOVERY", "HIGH");
end

package rules.cvm.offer

rule "Retention discount needs approval above threshold"
when
  $f : CvmOfferFact(offerType == "RETENTION_DISCOUNT",
                  discountPercent > 10)
  $d : CvmDecision()
then
  $d.requireApproval("CVM_HIGH_VALUE_RETENTION_OFFER");
end
```

7. Events

Event	When emitted	Key fields
CvmCustomerEvaluated	Signal/risk evaluation completed.	customerId, churnRiskScore, segments
CvmActivityCreated	Activity created.	activityId, activityType, customerId
CvmOfferProposed	Offer proposed.	offerInstanceId, offerType
CvmOfferAccepted	Customer accepts.	offerInstanceId, customerId
CvmOfferApplied	Owning module applied offer.	owningModuleRefType, owningModuleRefId
CvmActivityClosed	Activity closed.	activityId, outcomeCode

8. Implementation Notes

- Start with rules-based segmentation before ML scoring.
- Keep score explanations visible; do not create unexplained retention actions.
- CVM workers can be consolidated with reporting/event consumers if volume is low.
- All customer contact attempts must write to CUST-INT-01.
- Use REP-01 for dashboards; EM-03 should not build its own BI store.

9. Test Scenarios

1. Dunning event updates signal profile.
2. High-risk customer enters retention segment.
3. CVM activity is idempotent by source event.
4. Offer over threshold calls EM-CFG-04.
5. Accepted discount offer calls SIP-03, not DIS-OP directly.
6. Notification request goes through NOT-01.
7. Interaction is written to CUST-INT-01.